

Tax Productivity and Development: A Quantitative Macroeconomic Analysis

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The Importance of Tax Collection

- Core task of government; prerequisite for any public spending
- Central to 'state capacity' theory of development
 - Besley and Persson, (2011), Besley, Ilizetzi, and Persson (2013), Mayshar, Moav, and Pascali (2022), Tilly (1990) others

Why Was Egypt a Great State?



Egyptian Pyramids $\approx$ 3000 BCE



New Guinea sweet potatoes $\approx$ 5000 BCE

Answer: The Nilometer. Mayshar, Moav, Pascali (2021)'s Appropriability



Challenges Faced in the Literature

- Tax collection capacity or not easy to measure
 - Typically proxied by Taxes/GDP but this is quite indirect
 - Theories of state capacity $\Rightarrow$ tax capacity increases Taxes and GDP
- Little consensus on *quantitative importance* of tax collection capacity
 - Few answers to: How much would GDP / capita increase in the long-run if we doubled tax collection capacity in low-income African country?

Overview of This Paper

- New data on tax collection across 98 countries
 - Tax collections / collection expenditures
 - Document strongly positive correlation with GDP / capita in cross-section and country time series

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- Dynamic G.E. model to interpret data + simulate counterfactuals
 - Quantitative Ramsey model with tax administration, modern, and traditional sectors
 - Define notion of tax productivity
 - Production function of tax administration: $\tau = G(A^C, L_C)$
 - $A^C = \text{tax productivity}$, $L_C = \text{number of collectors}$, τ tax rate on modern sector wage bill,
 - Public capital and private capital complements in production a la Baxter-King (1993)
 - Lower collection capacity $\Rightarrow$ lower public capital $\Rightarrow$ lower GDP per capita

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- (Goal) Infer differences in tax productivity across countries
- (Today) Counterfactually doubling Sub-Saharan Africa's tax productivity would raise GDP per capita in the long run by $\sim 9\%$
 - Order of magnitude larger than simple "Hulten" logic, which suggests 0.9%

Empirical Findings: Tax Collections / Expenditures

Measuring Tax Collections/ Expenditure

- Unit of Observation: tax administration

Measuring Tax Collections/ Expenditure

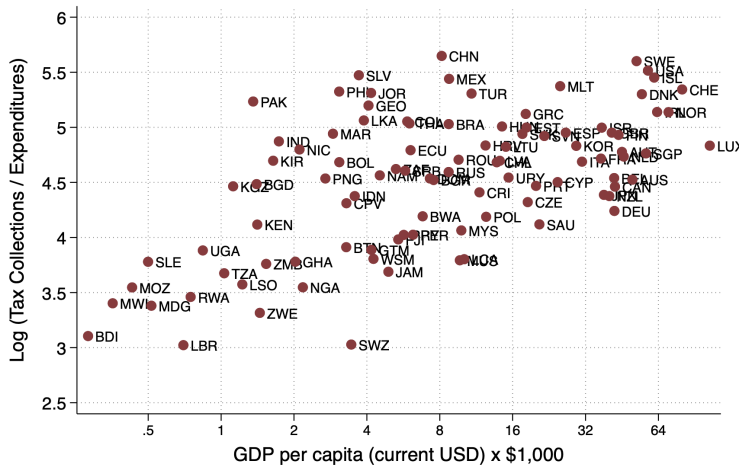
- Unit of Observation: tax administration



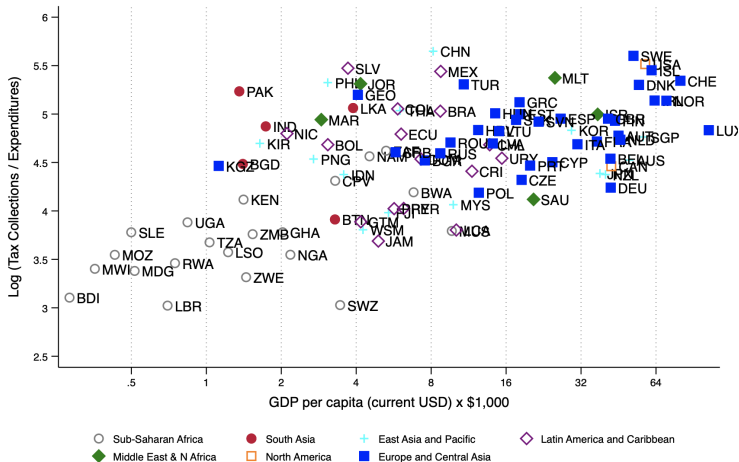
Measuring Tax Collection / Expenditure

- Data Source
 - National tax authority documents
 - IMF Revenue Administration Fiscal Information Tool
- Unit of Observation: tax administration
- 98 countries, 625 country-year observations
- T = Tax Collections including personal income and corporate taxes, VAT, sales and custom taxes
- X = Official tax collection authority budget
- Tax collection over collection expenditure, T/X

Tax Collection / Expenditure Increases with GDP per capita



Tax Collection / Expenditure Increases with GDP per Capita

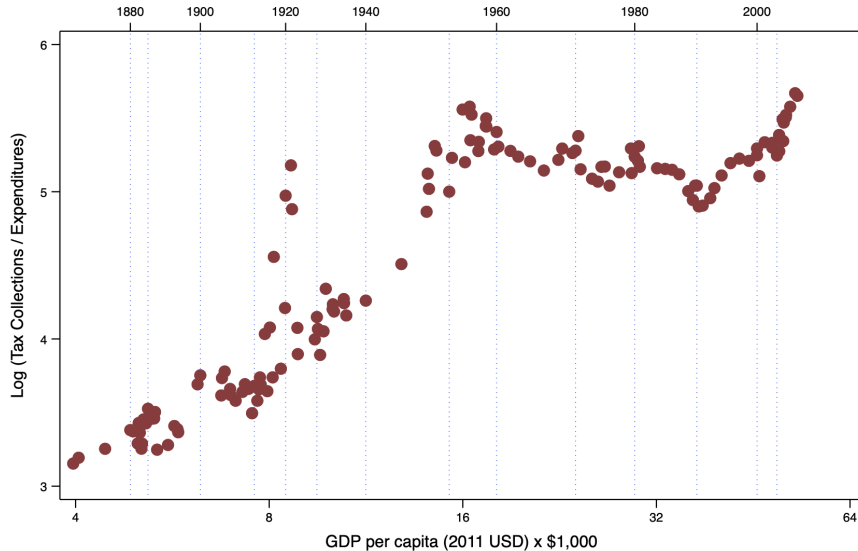


Descriptive Regressions: Tax Collection / Expenditure & GDP Per Capita

	(1) Log (T/X)	(2) Log (T/X) Exclude if not responsible for VAT/sales taxes	(3) Log (T/X) Exclude if not responsible for customs duties	(4) Log (T/X) Exclude if responsible for lotteries	(5) Log (T/X) Exclude if responsible for social security contributions
Log GDP pc	0.256*** (0.0363)	0.257*** (0.0367)	0.268*** (0.0367)	0.258*** (0.0362)	0.222*** (0.0443)
Countries	98	95	88	96	72
R-squared	0.341	0.346	0.384	0.351	0.264

Note: Each column represents one regression of log tax productivity, defined as taxes collected divided by tax administration expenditures, on log GDP per capita. *** means significant at 1%, ** means 5%, and * means 10%

Tax Collection / Expenditure in the US: 1867 to 2017



Descriptive Regressions: Historical Time Series

	(1) Log (T/X)	(2) Log (T/X) Exclude if not responsible for customs duties	(3) Log (T/X) Exclude if responsible for social security contributions	(4) Log (T/X) Exclude if changes in responsibilities
Log Real GDP pc	0.637*** (0.175)	0.636*** (0.197)	0.684*** (0.163)	0.644*** (0.193)
Observations	625	546	519	578
R-squared	0.563	0.541	0.574	0.541
Country FE	yes	yes	yes	yes

Note: Each column represents one regression of log tax productivity on on log GDP per capita with country fixed effects. the sample consists of 6 OECD countries, 1 from Latin America, 4 from Asia, and 5 from Sub-Saharan Africa. *** means significant at 1%, ** means 5%, and * means 10%

Model of Tax Productivity and Development

Overview of Model

- Positive correlation between collections / expenditure and GDP / capita suggestive of higher tax capacity in rich countries, but not directly interpretable
- Need structural model to:
 - Formalize notion of **tax productivity**
 - Measure collection capacity in cross section of countries
 - Simulate effects of counterfactually raising collection capacity

Overview of Model Environment

- Ramsey model à la Chamley (1986)
- Public and private capital in the production function (Baxter-King, 1993)
- Modern and Traditional production sectors
- Novelty: Tax administration sector
 - One input: tax collectors
 - Output: tax rate on modern sector wages
 - Tax productivity is exogenous, captures “tax collection capacity”

Technology: Modern and Traditional Sectors

- Modern firms production function:

$$Y_M = F(A^M, k_G, k, L_M),$$

- k_G : public capital,

Firms hire workers at wage $w_{M,t}$ and rent capital at rate r_t

$$\max_{k_t, L_{M,t}} F(A^M, k_{G,t}, k_t, L_{M,t}) - w_{M,t} L_{M,t} - r_t k_t.$$

- **Traditional Sector:**

Firms in the traditional sector have a linear production function:

$$Y_T = A^T L_T,$$

Workers are hired at a fixed wage $w_T = A^T$.

Public Sector and Government

Tax Collection:

Taxes T_t are collected on modern sectors wages at rate $G(A^C, L_{C,t}) \leq 1$,

$$T_t = G(A^C, L_{C,t}) w_{M,t} L_{M,t}.$$

Government budget: can borrow to finance budget.

$$w_{M,t} L_{C,t} + R_t b_t + k_{t+1}^G \leq T_t + b_{t+1} + (1 - \delta^G) k_t^G.$$

Public capital evolves as:

$$k_{G,t+1} = (1 - \delta^G) k_{G,t} + T_t.$$

Government cannot borrow to finance public capital.

Households

Modern and traditional sector workers form a representative household that owns all capital and supplies labor. The household maximizes its lifetime utility:

$$\max_{\{c_t, k_{t+1}, b_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t, L_{M,t}, L_{T,t}),$$

$$\text{subject to: } c_t + k_{t+1} + b_{t+1} \leq (1 + r_t - \delta)k_t + (1 - G(A^C, L_{C,t}))w_{M,t}L_{M,t} \\ + w_T L_{T,t} + R_t b_t.$$

Initial private capital $k_0 = \bar{k}_0$.

Equilibrium

A competitive equilibrium in this economy consists of allocations

$$\{c_t, L_{T,t}, L_{M,t}, L_{C,t}, k_t, k_{G,t}, b_t\}$$

and prices

$$\{w_{T,t}, w_{M,t}, r_t, R_t\},$$

such that:

1. Firm Optimality:

- Modern firms take prices and $k_{G,t}$ as given, and maximize profit.
- Traditional firms hire labor at wage $w_T = A^T$.

2. Household Optimality: The household maximize utility for given prices and tax collection personnel.

3. Budget constraints: Government budget constraint and public capital accumulation eqs hold.

4. Market Clearing: The goods market clears:

$$c_t + k_{t+1} + w_{M,t}L_{C,t} + k_{G,t+1} \leq (1 + r_t - \delta)k_t + w_{M,t}L_{M,t} + A^T L_{T,t} + (1 - \delta^G)k_{G,t}.$$

Planner's Problem

To study optimal policy, the planner (or tax authority) chooses the sequence of tax collection personnel $\pi = \{L_{C,t}\}_{t=1,\dots,\infty}$ to maximize the household's lifetime welfare:

$$\max_{\pi} \sum_{t=0}^{\infty} \beta^t u(c_t(\pi), L_{M,t}(\pi) + L_{T,t}(\pi)),$$

subject to:

- the aggregate resource constraint,
- an **implementability condition** that ensures allocations are consistent with household optimization, and
- the government's budget constraint.

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Lemma

The implementability condition on the planner's allocation choice is:

$$\sum_{t=0}^{\infty} \beta^t \left[u_c(t) c_t - u_{L_M}(t) L_{M,t} - u_{L_T}(t) L_{T,t} \right] = u_c(0) R_0 k_0.$$

Optimal Tax Collection

At an interior steady state, the optimal tax collection choice satisfies:

$$G_{L_C}(A^C, L_C) = (1 - (1 - \delta_G)\beta) \frac{1}{L_M} \frac{F_k + 1 - \delta}{(F_G - \delta_G) - (F_k - \delta)}.$$

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Economic Interpretation:

- A high F_k (marginal return on private capital) favors private investment over public hiring.
- A high F_G (marginal return on public capital) enhances the attractiveness of public sector spending.
- A higher tax collection productivity A^C improves the effectiveness of tax collectors.
- A larger modern sector (L_M) expands the taxable base.

All endogenous object.

Transition Dynamics

Lemma

Assume that household preferences are separable in consumption and labor, with CRRA utility in consumption, and that the tax production function is linear in personnel $G(A^C, L_C) = A^C L_C$. Then an interior steady state exists and the optimal personnel choice over the transition (for $t > 0$) satisfies:

$$\frac{(F_G(t+1) - \delta_G) - (F_k(t+1) - \delta)}{F_k(t+1) + 1 - \delta} = \frac{1}{A^C} \left(\frac{1}{L_{M,t}} - \frac{\beta(1 - \delta_G)}{L_{M,t+1}} \right).$$

Higher tax productivity A^C leads to higher levels of public capital (lower F_G).

Quantitative Analysis

Quantitative Analysis

- Goal:

- Infer tax productivity A^C across countries
- Simulate long-run aggregate effects of raising collection capacity
- Simulate transition dynamics

- Today:

- Proof of concept calibration
- Long effects of doubling tax productivity in Ghana

Functional Forms and the Steady State

- We set the following functional forms:

$$G(A^C, L_C) = A^C L_C^\eta,$$

$$F_M(k^G, k, L_M) = A_M (k^G)^\zeta k^\mu L_M^{1-\mu},$$

$$F_T(L_T) = A_T L_T,$$

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma},$$

$$u(L) = \frac{(L)^{1/\epsilon}}{1 + 1/\epsilon}.$$

- We introduce private capital wedges $\mu_{k,t}$ in the household's capital decision to capture:
 - realistic distortions in emerging markets
 - potential distortionary effects of public capital.

Steady State

A steady state is determined by the following system of 9 equations in the 9 unknowns $(w_M, w_T, r, L_M, L_T, L_C, k, k_G, c)$:

(i) Optimal collection:
$$\frac{1 - \beta(1 - \delta_G)}{G_{L_C}(A^C, L_C) L_M} = \beta [F_G + (1 - \delta_G)] - 1 + \beta \left(\frac{1 - \eta}{\eta} \right) F_{L_M G} L_C,$$

(ii) Private capital:
$$1 = \beta (F_k + 1 - \delta) + \beta \frac{1 - \eta}{\eta} F_{L_M, k} L_C,$$

(iii) Public capital:
$$k_G = \frac{1}{\delta^G} G(A^C, L_C) w_M L_M,$$

(iv) Return on private capital:
$$r = F_{M, k},$$

(v) Modern sector wage:
$$w_M = F_{M, L},$$

(vi) Traditional sector wage:
$$w_T = F_{T, L},$$

(vii) Modern labor supply:
$$u_{L_M} = u_c w_M (1 - G(A^C, L_C)),$$

(viii) Traditional labor supply:
$$u_{L_T} = u_c w_T,$$

(ix) Aggregate resource constraint:
$$c + k + w_M L_C + k_G = (1 + r - \delta)k + w_M L_M + A^T L_T.$$

Experiment: Double Tax Productivity (A^C) of Ghana



See also Jawara, Knebelmann, Levine, Ndiaye, and Pouliquen (2025) [In the field. The Gambia]

Impact of Doubling Tax Productivity

Parameter	Baseline	Counterfactual ($A^C \times 2$)
GDP		+8.6%
Wages (Modern)		+6%
Public Capital		+75%
Private Capital		+14%
Tax collection / Expenditure	77	176
Effective Tax Rate	2.5%	3.9%
Taxes to GDP	1%	1.6%
Employment share in tax collection	1.5%	1.1%

- Domar weight of public sector: $w_M L_C / GDP = 0.88\%$
- Hulten's theorem would predict GDP $+0.88\% \times 100\% = +0.88\%$
- We find GDP growth of an order of magnitude larger (+8.6%)

Conclusion and Future Work

- New measures of tax collection capacity across countries
- Richest countries 30 more collection capacity than poorest
- Disciplined by new data of tax collection per expenditure on collection
- Quantitative Ramsey model to study tax productivity and development
- Long-run effects of raising capacity could be substantial – much larger than those suggested by simple Hulten calculation
- Future work: fine-tuned calibrations to several countries, inference of tax productivity differences across countries, transition dynamics, alternative elasticities of substitution between public and private capital, **your ideas**.

Thanks!



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